A quadratic speed bound for two-entry modifications of the Lonely Runner baseline

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Abstract

Let $n \geq 5$, remove two distinct integers from $1, \ldots, n-1$, and insert two integers $n \leq w_1 < w_2$. We prove that if the resulting unshifted runner set covers the circle at radius $1/n$, then $w_2 < 28n^2$. Thus every tight two-entry modification belongs to an explicit finite search domain, without a restriction on the removed speeds or a primitivity assumption. The proof first shows, by elementary interval arguments, that two deletions followed by one insertion always leave a strict hole. Rational endpoints and periodic occupancy then control the inserted speeds. For upper-half deletions, a unit residue at a removed-speed center gives a stronger arithmetic bound; a separate argument handles an insertion divisible by both removed speeds. The middle windows come from Zhang's one-entry analysis, and the unit-gap estimate is due to Goddyn and Wong. All necessary ingredients are proved here. Exact computations subsequently classify this family through $n = 30$. The universal quadratic theorem is a paper proof; the computational and formal evidence have separate, explicitly limited scopes.

1. Statement and relation to prior work

For a finite nonempty set V of positive integers, put

$$\mathrm{LR}(V) = \max_{t \in [0,1]} \min_{v \in V} \|vt\|, \qquad \|x\| = \min_{k \in \mathbb{Z}} |x - k|.$$

We call V a cover at radius $1/n$ if

$$\forall t \in \mathbb{R} \quad \exists v \in V : \quad \|vt\| \leq 1/n.$$

By continuity and periodicity this is equivalent to $\mathrm{LR}(V) \leq 1/n$. Tightness means equality. Throughout the paper, all starting shifts are zero, and $[n-1] = \{1, \ldots, n-1\}$.

Theorem 1. Let $n \geq 5$, $1 \leq r_1 < r_2 \leq n-1$, and $n \leq w_1 < w_2$ be integers. Set

$$S = [n-1] \setminus \{r_1, r_2\}, \qquad V = S \cup \{w_1, w_2\}.$$

If $\mathrm{LR}(V) \leq 1/n$, then

$$w_2 < 28n^2. \tag{1}$$

When $2r_1 \le n-1$, the more precise bounds

$$w_1 < 14n, \qquad w_2 < 28n(n-1) \tag{2}$$

hold. When both removals are upper-half and $n \ge 7$, we obtain $w_1 < 2n^2$ and $w_2 < 12n^2$.

Zhang [1] classifies one-entry modifications and asks for polynomial growth bounds for more general modifications. His interval windows supply an important part of our proof. We formulate their consequences directly for coverage, rather than inferring a coverage theorem from a statement about equality. Goddyn and Wong [2] proved the elementary unit-gap bound used below and classified multiplicative one-entry modifications. Liu [3] determined the corresponding one-deletion values; here only a directly proved modular lower witness is needed.

A previously accepted formal proof [4] gives a stronger classification when both removals are upper-half. Its interfaces were inspected but its complete proof was not replayed locally. The present proof does not depend on it. General geometric finite-checking results [5,6] provide broader context; they do not directly yield the unconditional bound (1) for this source family. Academic priority of the unrestricted consequence in Theorem 1 remains under review. No general Lonely Runner Conjecture or classification for all n is asserted.

2. Two elementary tools

Write $B_w = \{t : \|wt\| \le 1/n\}$. Its connected components on the real line are the disjoint closed intervals

$$\left[\frac{k}{w} - \frac{1}{nw}, \frac{k}{w} + \frac{1}{nw}\right], \qquad k \in \mathbb{Z}. \tag{3}$$

Every connected interval contained in B_w lies in one such band. If all retained speeds are strictly outside their bad sets on an interval, coverage forces the inserted speeds to cover that interval.

Lemma 2 (unit gaps). Every interval of $\lfloor r/2 \rfloor + 1$ consecutive integers contains an integer coprime to $r \ge 2$. If $r \ge 7$, the least such guaranteed length $j(r)$ satisfies $j(r) \le r/2 - 1$.

Proof. For odd $r = 2q+1$, consider the units $1, q, q+1, 2q$. For $r = 4h$, use $1, 2h-1, 2h+1, 4h-1$. For $r = 4h+2 \ge 6$, use $1, 2h-1, 2h+3, 4h+1$. Discard repetitions. Their maximum cyclic gaps are respectively bounded by

$$\max(q-1, 2), \qquad \max(2h-2, 2), \qquad \max(2h-2, 4).$$

The listed integers are coprime to r; in the even cases their gcd with r divides 2 or 4 and they are odd. The asserted estimates follow. For $r = 2$ the unique unit has cyclic gap 2. Between consecutive listed units there are gap-minus-one consecutive nonunits. □

The first assertion is Goddyn–Wong's Lemma 2.1 [2]. The explicit unit construction above replaces their use of Bertrand's postulate. We will also use the exact small values

$$j(1), \dots, j(6) = 1, 2, 2, 2, 2, 4. \tag{4}$$

Lemma 3 (periodic occupancy). For every real interval J of length ℓ and positive integer w,

$$|J \cap B_w| \le \frac{2\ell}{n} + \frac{2(n-2)}{n^2 w}. \tag{5}$$

Proof. Put $\alpha = 2/n$ and write $w\ell = m + u$, where m is an integer and $0 \le u < 1$. Full periods contribute $m\alpha$; the remaining arc meets the bad arc in measure at most $\min(u, \alpha)$. Its excess over αu is at most $\alpha(1-\alpha)$. Divide by w. A single complete bad band attains this additive constant. □

If two distinct speeds $w_1 < w_2$ cover a retained-safe interval, (5) gives

$$\ell(1 - 4/n) < \frac{4(n-2)}{n^2 w_1}. \tag{6}$$

The strict inequality uses $w_2 > w_1$. The restriction $n > 4$ allows division by $1 - 4/n$.

3. Two deletions and one insertion leave a hole

Proposition 4. For every $n \ge 5$, distinct deletions $1 \le r_1 < r_2 < n$, and integer $w \ge n$,

$$\mathrm{LR}(([n-1] \setminus \{r_1, r_2\}) \cup \{w\}) > 1/n. \tag{7}$$

We prove this without using a one-entry equality classification. The next lemma reconstructs the middle-window argument of [1] under coverage alone.

Lemma 5 (middle removal). If $3 \le r \le (n-1)/2$ and $w \ge n$, then $([n-1] \setminus \{r\}) \cup \{w\}$ is not a cover.

Proof. Put $s = n - r$. For nonzero residues $x \pmod r$, let $F(x)$ be the representative of x in $[s, n-1]$. Let

$$I = \min\{i \ge s : \gcd(i, r) = 1\} = s + d.$$

Thus $I \le n - 1 < 2s$, and Lemma 2 gives $d \le \lfloor r/2 \rfloor$. For a unit $p \pmod r$, write $a = p^{-1} \pmod r$. All retained speeds are safe on each of the intervals

$$L_p = \left(\frac{p}{r} - \frac{s}{rnF(a)}, \frac{p}{r} - \frac{1}{2rn}\right), \qquad R_p = \left(\frac{p}{r} + \frac{1}{2rn}, \frac{p}{r} + \frac{s}{rnF(-a)}\right). \tag{8}$$

To verify this source statement, write a time in R_p as $p/r + u$. Then $1/(2rn) < u < 1/(rn)$. Every retained multiple kr, $k \ge 2$, has distance $kru > 1/n$, while $kru < 1/r \le 1/3$. A nonmultiple whose residue after multiplication by p is $r - 1$ has distance $1/r - iu > 1/n$, because $i \le F(-a)$. Every other nonzero residue stays strictly between $1/r$ and $(r-1)/r$ after the positive displacement. Reflection proves the left-window statement. These intervals are nonempty since all their F-values are below $2s$.

One of the right windows has length

$$\frac{2s-I}{2rnI}. \tag{9}$$

We need the inequality

$$\frac{4rI}{2s-I} < r(n-2). \tag{10}$$

It is equivalent to $d < s(n-6)/(n+2)$. If $r \geq 6$, the right side is at least $(r+1)(2r-5)/(2r+3) > r/2 \geq d$. The strict step is equivalent to $2r^2 - 9r - 10 = (r-6)(2r+3) + 8 > 0$. For $r = 4, 5$, (4) gives $d \leq 1$, while the right side is already $15/11$ and $30/13$, respectively, at the smallest admissible n. For $r = 3$, $d = 0$ at $n = 7, 8$, and for $n \geq 9$ the right side is at least $18/11 > d$. This proves (10).

Suppose there were a cover. Each window and its closure must lie in one w-band. Comparing (9) with its length $2/(nw)$ gives $w \leq 4rI/(2s-I) < r(n-2)$. Write $w = Kr + c$, with $-r/2 < c \leq r/2$, and put $F_+ = F(1)$, $F_- = F(-1)$. Covering the closed right window at $p = 1$ by a band centered at j/w requires

$$\frac{c}{r} + \frac{ws}{rnF_-} - \frac{1}{n} \leq j - K \leq \frac{c}{r} + \frac{w}{2rn} + \frac{1}{n}. \tag{11}$$

If $j - K \geq 1$, the right inequality forces $c \geq r - w/(2n) - r/n > r/2$. If $j - K \leq -1$, the left inequality forces $c < -r/2$. Thus $j - K = 0$, giving

$$c \leq \frac{r}{n} - \frac{ws}{nF_-} < 0. \tag{12}$$

The final sign follows from $ws \geq ns > r(n-1) \geq rF_-$. For the left window the analogous bounds are

$$\frac{c}{r} - \frac{w}{2rn} - \frac{1}{n} \leq j - K \leq \frac{c}{r} - \frac{ws}{rnF_+} + \frac{1}{n}.$$

Again the integer is zero, and hence $c \geq ws/(nF_+) - r/n > 0$, contradicting (12). □

Lemma 6 (small removals). A one-entry cover with $r = 1$ is impossible. For $r = 2 \leq (n-1)/2$, coverage is possible only when $(n, w) = (5, 7)$ or $(6, 9)$.

Proof. For $r = 1$, the interval $(1/(2n), 1/n)$ is safe for all speeds $2, \ldots, n-1$. Its length $1/(2n)$ exceeds $2/(nw)$ for $w \geq n \geq 5$, so it cannot fit in one w-band. This argument already appears in [2].

For $r = 2$, the left shell

$$(1/2 - \beta, 1/2 - 1/(4n)), \quad \beta = \begin{cases} 1/(2n) & n \text{ odd}, \\ (n-2)/(2n(n-1)) & n \text{ even} \end{cases} \tag{13}$$

is safe. Indeed, at $t = 1/2 - u$, retained even speeds have distance $vu > 1/n$, with $vu < 1/2$; retained odd speeds have distance $1/2 - vu > 1/n$, using the largest odd speed to check the last inequality.

A covering w-band has its center below $1/2$: its halfwidth $1/(nw) \le 1/n^2 < 1/(4n)$ is smaller than the shell's distance from $1/2$. The largest such center is $1/2 - 1/w$ for even w, and $1/2 - 1/(2w)$ for odd w. Reaching the right endpoint of the closed shell forces respectively $w \ge 4(n-1)$ or $w \ge 2(n-2)$. The shell length gives $w \le 8$ for odd n and $w \le 8(n-1)/(n-3)$ for even n. Even w is impossible. For odd w, these inequalities leave only $(5,7), (6,9), (6,11), (6,13)$. At $t = 87/200$, the distances for speeds $1, 3, 4, 5, 11, 13$ are $87, 61, 52, 35, 43, 69$, divided by 200; all exceed $1/6$. Thus the last two cases fail. This is the shell argument of [1]. □

Proof of Proposition 4. If the two-deletion, one-insertion set covered, adding either missing speed would give two one-entry covers. Lemmas 5–6 exclude every lower-half removal except 2 in the two sporadic cases. For $n = 5, w = 7$, the other removal is 3 or 4, and its center $1/r$ is a hole since $r \nmid 7$. For $n = 6, w = 9$, centers exclude other removals 4 and 5. If the pair is $\{2, 3\}$, at $t = 13/36$ the speeds $\{1, 4, 5, 9\}$ have distances $13, 16, 7, 9$, divided by 36, all above $1/6$.

It remains to consider two upper-half removals. At $1/r_i$, every retained speed has distance at least $1/r_i > 1/n$. If $r_i \nmid w$, the inserted speed also does, giving a hole. Otherwise write $w = mr_1 = lr_2$, with $m > l \ge 1$. Then

$$mr_1 = lr_2 \le (m-1)(n-1), \qquad m(n-r_1) \ge n - 1 + m \ge n + 1.$$

Take $u = (n+1)/(n^2 w)$. The preceding inequality gives $u < (n - r_1)/(nr_1(n-1))$. At $t = 1/r_1 + u$, all retained speeds remain strictly above $1/r_1 - (n-1)u > 1/n$. The inserted distance is determined by $(n+1)/n^2 \in (1/n, 1 - 1/n)$, so it too is safe. This contradiction exhausts all cases. □

4. Endpoint spacing and lower-half removals

Lemma 7. Under the cover hypothesis of Theorem 1,

$$w_2 \le 2(n-1)w_1. \tag{14}$$

Proof. Proposition 4 gives a nonempty open gap after inserting w_1. Both endpoints are actual endpoints $(nk \pm 1)/(nv)$ of source bands. There are no exceptional endpoints at 0 or 1, which have bad neighborhoods. For different endpoint labels u, v, their positive difference is an integer divided by nuv. At most one label exceeds $n-1$, so the gap has length at least $1/(nw_1(n-1))$. For equal labels its length is an integer divided by nv, giving an even larger lower bound. The gap must fit in a single w_2-band of length $2/(nw_2)$. □

If the retained minimum has value at least $\mu > 1/n$ at some time, its Lipschitz constant at most $n-1$ gives a safe interval of length $2(\mu - 1/n)/(n-1)$. Equation (6) yields

$$w_1 < \frac{2(n-2)(n-1)}{n(n-4)(\mu - 1/n)}. \tag{15}$$

Proposition 8. If $2r_1 \le n - 1$, then (2) holds.

Proof. Let D be the least integer at least $n + r_1$ coprime to r_1. Choose $ar_1 = 1 \pmod D$. At a/D, every retained speed has distance at least $2/D$: residues 0 and ± 1 would require respectively a multiple of D, the deleted speed r_1, or a speed at least $D - r_1 \ge n$. Thus we may use $\mu = 2/D$ in (15).

For $r_1 \geq 7$, Lemma 2 gives

$$D \leq n + r_1 + j(r_1) - 1 \leq n + 3r_1/2 - 2 \leq (7n - 11)/4. \tag{16}$$

Using (4), the same estimate holds for smaller r_1, except at the five pairs in the following table. The final column is (15).

n, r_1	D	strict bound for w_1
$5, 2$	7	56
$6, 2$	9	60
$7, 3$	10	50
$10, 4$	15	72
$14, 6$	23	3588/25

This list is exhaustive without an unbounded computation: the bound $D \leq n + r_1 + j(r_1) - 1$ implies (16) for $n \geq 5, 8, 9, 11, 12, 16$, respectively, when $r_1 = 1, \ldots, 6$. Only the intervening admissible integers $n \geq 2r_1 + 1$ remain, and their gcds give the displayed exceptions. Each table entry is below $14n$.

Otherwise, (16) gives $\mu - 1/n \geq (n + 11)/(n(7n - 11))$. The bound (15) is below $14n$, since the required positive numerator is

$$\begin{aligned} 7n(n-4)(n+11) - (n-2)(n-1)(7n-11) &= 81n^2 - 355n + 22 \\ &= 81(n-5)^2 + 455(n-5) + 272 > 0. \end{aligned}$$

Thus $w_1 < 14n$. Lemma 7 gives $w_2 < 28n(n-1)$. □

5. Arithmetic centers for upper-half removals

Assume $2r_1 > n - 1$. The safe center $1/r_1$, used in (15), gives

$$w_1 < \frac{2(n-2)r_1(n-1)}{(n-4)(n-r_1)} \leq \frac{(n-2)^2(n-1)}{n-4} < 2n^2. \tag{17}$$

Here $r_1 \leq n - 2$. The last inequality follows from

$$2n^2(n-4) - (n-2)^2(n-1) = (n-5)(n^2 + 2n + 2) + 14 > 0.$$

Lemma 9 (a unit center). If r, a are positive integers and $r \nmid a$, some integer p coprime to r satisfies $\|ap/r\| \geq 1/6$.

Proof. Put $g = \gcd(a, r)$, $q = r/g > 1$. A unit $b \pmod q$ at distance at least $1/6$ from an integer after division by q is explicit:

$$b = \begin{cases} 1, & q = 2, \\ (q-1)/2, & q \text{ odd}, \\ q/2 - 1, & q \equiv 0 \pmod 4, \\ q/2 - 2, & q \equiv 2 \pmod 4,\ q \geq 6. \end{cases}$$

The corresponding distances are $1/2$, $1/2 - 1/(2q)$, $1/2 - 1/q$, and $1/2 - 2/q$. Coprimality is immediate; the minimum $1/6$ occurs at $q = 6$. Reduction of units modulo r onto units modulo its divisor q is surjective: prescribe a unit modulo q and prescribe 1 modulo each prime dividing r but not q, then apply the Chinese remainder theorem. Lift $b(a/g)^{-1} \pmod q$ to such a unit $p \pmod r$. Then $ap/r = b/q \pmod 1$. $\square$

Suppose first that $n \geq 7$ and some removed $r \nmid w_1$. Choose the unit p of Lemma 9 with $0 < p < r$, and put $t_0 = p/r$. All retained baseline distances there are at least $1/r$. Consequently the interval of radius

$$R = \min\left(\frac{n-r}{nr(n-1)}, \frac{n-6}{6nw_1}\right) > 0 \tag{18}$$

around t_0 is safe for both the retained baseline and w_1. It stays inside $(0, 1)$, since $R < 1/r$. At the center, final coverage forces $r \mid w_2$, as a nonzero circle distance with denominator $r < n$ exceeds $1/n$. The entire connected safe interval must lie in the w_2-band centered exactly at t_0. Hence

$$w_2 \leq \max\left(\frac{r(n-1)}{n-r}, \frac{6w_1}{n-6}\right) < 12n^2, \tag{19}$$

using (17) and $n \geq 7$.

It remains possible that w_1 is divisible by both removals. Write $w_1 = mr_1 = lr_2$, $m > l$, so $m(n - r_1) \geq n - 1 + m$. Define

$$A = \frac{n - r_1}{nr_1(n-1)}, \qquad h = \frac{1}{nw_1}.$$

The interval

$$\left(\frac{1}{r_1} + h, \frac{1}{r_1} + \min\left(A, \frac{n-1}{nw_1}\right)\right) \tag{20}$$

is safe for the retained baseline and lies between consecutive w_1-bands. Its length is at least

$$\min\left(\frac{1}{nr_1(n-1)}, \frac{n-2}{nw_1}\right),$$

because

$$A - h = \frac{m(n - r_1) - (n-1)}{nw_1(n-1)} \geq \frac{1}{nr_1(n-1)}.$$

Fitting (20) in one w_2-band gives

$$w_2 \le \max\left(2r_1(n-1), \frac{2w_1}{n-2}\right) < 2n^2. \tag{21}$$

The next-band endpoint in (20) is essential even when w_1 is extremely large. No distinct assignment of insertions to removals has been assumed.

Completion of Theorem 1. Proposition 8 handles all pairs with a lower-half removal. Equations (19)–(21) handle two upper-half removals for $n \ge 7$. For $n = 5, 6$, (14) and (17) give $w_2 < 4(n-1)n^2 \le 20n^2 < 28n^2$. These cases exhaust the domain. □

6. Finite reduction and computational consequences

Corollary 10. For each fixed $n \ge 5$, every two-entry cover is found by testing the finite set of integers

$$1 \le r_1 < r_2 < n, \qquad n \le w_1 < w_2 \le 28n^2 - 1.$$

Stronger parameter bounds from (14)–(19) can reduce this domain. Exact coverage is decided by the finite union of closed source bands (3) on $[0, 1]$; integer cross-products suffice for all endpoint comparisons. This finite interval method is established, not a new certification principle.

Two separately implemented exact computations classify the family through $n = 30$. The only covers are

n	$\{r_1, r_2\}$	$\{w_1, w_2\}$	V
5	$\{1, 3\}$	$\{6, 8\}$	$\{2, 4, 6, 8\}$
8	$\{2, 3\}$	$\{11, 13\}$	$\{1, 4, 5, 6, 7, 11, 13\}$.

Both have loneliness exactly $1/n$: exact coverage supplies the upper bound, and evaluation at $t = 1/n$ supplies the lower bound. The first has gcd 2 and must not be removed by a primitive-only filter. The second is a known Goddyn–Wong example [2], not a new construction.

The executed searches used the earlier proved parameter-dependent cubic domains, which contain all possibilities and remain valid after Theorem 1. For $5 \le n \le 20$, those domains contained 79,648,374 quadruples. The extension $21 \le n \le 30$ covered 923,839,088 quadruples with no survivors; exact longest-gap, width, and center-divisibility pruning reduced the individual tests. A separate Fraction implementation, importing no SUCC search or verifier modules, agreed. The mathematical bound is proved above; agreement of implementations is not its proof. Reproducible receipts and the complete pruning derivations accompany the repository dossiers.

Appendix. Verification and limitations

The paper proof preserves zero shifts, the actual baseline, all deletion pairs, and nonprimitive speed sets. Positive controls include the established two-acceleration examples at $n = 74$ and $n = 284$. Negative controls include wrong divisibility assignments, omitted coverage gaps, missing periodic boundary slack, and strict-to-integer rounding errors. The unit-center constant cannot be increased above $1/6$ without additional hypotheses, as modulus 6 shows.

Lean evidence and executable evidence must be read by their actual scopes. The earlier checked development includes a complete rational witness for the upper-half one-insertion hole and generic source-band coverage semantics. The new growth formalization proves the complete upper-half implication $w_2 < 3n^3$ from actual coverage, including source-gap construction and the two-band interval estimate. Its 18 exported theorem types, standard-axiom receipts and three paired false controls delimit this authority. No finite scan, a priori separation certificate, or earlier upper-half theorem is advertised as a kernel proof of the unrestricted quadratic theorem. Theorem 1 remains a paper proof. Same-model internal reviews are correlated checks, not external peer review.

The constant 28 is not asserted optimal. An $O(n \log n)$ bound for the unrestricted family, a classification for all n, and growth estimates for three or more modifications are not established here.

References

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